Postpartum Hemorrhage

Administer High Flow Oxygen
10-15lpm via NRB Mask

IV Access
X2 Large Bore IV line above the diaphragm

Consider IV Normal Saline
500ml Boluses
If hemodynamically unstable
(max 2000ml)

Assess Perineum for trauma
Provide direct pressure to control any trauma/tears where needed

Placenta Delivered?

NO

Complete 3rd Stage of Labor

Hemorrhage Controlled?

YES

YES

- **Fundal Massage**
- **Encourage mother to empty bladder**

Hemorrhage Controlled?

YES

NO

"Manage the T's"

TONE: Fundus firm and central?
- Continue fundal massage

TRAUMA: Evidence of perineum or vaginal wall lacerations?
- Control hemorrhage

TISSUE & THROMBIN: Placenta and membranes intact? Evidence of clots? Known clotting disorder or patient anticoagulated?
- Continue fundal massage to assist with expulsion of clots

Hemorrhage Controlled?

YES

NO

Consider:
- **Tranexamic acid:**
  - 1gram IV over 10 minutes
- **Sodium chloride**
  - 500ml boluses (max 2000ml)
- **External aortic compression**

Patient Care Goals

1. Recognize serious conditions associated with hemorrhage during pregnancy, even when the cause is not apparent (e.g., ectopic pregnancy, abruptio placenta, placenta previa).
2. Provide adequate resuscitation for hypovolemia and stabilize the patient.
3. Ensure rapid transport to an appropriate receiving facility to address potential fetal compromise and maternal shock.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Primary Post Partum Hemorrhage (PPH):
  - Blood loss >500 ml per vagina within the first 24 hours post-birth.
- Secondary Post Partum Hemorrhage (PPH):
  - Acute excessive bleeding between 24 hours and 12 weeks post-birth.
- Signs of worsening hypovolemia (altered consciousness, cold/clammy skin) following delivery.

Exclusion Criteria

- Childbirth and active labor (See Childbirth CPG)
- Antepartum hemorrhage (See PV Hemorrhage in Pregnancy CPG).

Key Patient Assessment Points

1. **Obstetric History**
   - Obtain obstetric history: gravity, parity, duration of labor, placenta delivery status, and any risk factors (e.g., multiple gestation, previous PPH, or coagulopathies).
2. **Physical Examination**
   - Uterine Assessment:
     - Fundal Height and Tone: Palpate the uterus to assess if it is atonic (soft, boggy) or contracted. Atony is the leading cause of PPH. Fundal height above the umbilicus may suggest retained products or clots.
   - Vaginal and Perineal Examination:
     - Inspect for lacerations, hematomas, or ongoing bleeding from the perineum, vagina, or cervix. Look for trauma-related bleeding, especially in cases of instrumental delivery.
3. **Cause Identification** (4 T's Mnemonic)
   - Tone: Assess for uterine atony (most common cause).
   - Trauma: Identify lacerations, hematomas, or uterine rupture.
   - Tissue: Look for retained placenta or clots (external visual observation only)
   - Thrombin: Consider coagulopathy (e.g., prolonged bleeding, petechiae, or history of clotting disorders).
4. **Estimation of Blood Loss & Bleeding Severity**
   - Quantify vaginal bleeding (e.g., number of soaked pads, passing clots >5 cm, or estimated loss >500 mL for vaginal delivery).
   - Assess the rate PV bleeding.
5. **Evaluate for Signs of Shock or Instability**
   - Assess for signs of shock:
     - Evaluate mental status (e.g., confusion, agitation, or lethargy).
     - Note pallor, sweating, or clammy skin, dizziness, weakness, or palpitations.
     - Assess heart rate (>100 bpm), or blood pressure (<90/60 mmHg) are indicative of significant blood loss.
     - Postural blood pressure assessment (A drop in BP >20 mmHg, and/or a rise in heart rate between lying and standing measurements indicates significant hypovolemia.
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - Ongoing PV bleeding with or without signs of shock.
       - Estimated blood loss >500mL

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
   - Determine obstetric history.
1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
2. **General Management**
   - Place absorbent pad to perineum.
   - Calm and reassure mother.
   - Ensure mother is in a position of comfort (left lateral tilt if possible).
   - Consider requesting ALS back-up.
3. **Administer high-flow oxygen**
   - At 10-15 Lpm via non-rebreather (NRB) mask.
4. **Establish large-bore IV access**.
   - Consider second large bore IV if patient is hemodynamically unstable.
5. **If Hemodynamically Unstable:**
   - Administer 500 mL boluses of Normal Saline as needed for hypotension, up to a maximum of 2000 mL, while monitoring perfusion and mental status.
6. **If Placenta Not Delivered:**
   - Assess the perineum for trauma and provide direct pressure if trauma/tears are present.
   - Await placental delivery (3rd stage of labor).
7. **If Placenta Delivered:**
   - Perform fundal massage and encourage the mother to empty her bladder.
8. **If Hemorrhage Still Uncontrolled:**
   - Ensure fundus is firm and central; continue fundal massage to assist with expulsion of clots.
9. **Consider Tranexamic acid (TXA):**
   - Indications:
     - Profuse bleeding, with signs of shock or/
     - Significant shock symptoms
   - Dose:1 gram IV, diluted in 100ml NaCl, administered over 10 minutes.
10. **External Aortic Compression**
    - Apply external aortic compression and do not release until the receiving facility takes over pressure.
    - External Aortic Compression Procedure:
      - Position the Patient:
        - Place the patient flat on her back (supine) on a firm surface, such as a stretcher or table.
      - Locate the Compression Point:
        - Identify the abdominal aorta, which is located just above the umbilicus (belly button) and slightly to the left of the midline. This is roughly at the level of the T12-L4 region.
      - Apply Firm Pressure:
        - Stand beside or slightly above the patient.
        - Using both hands or a closed fist, press down firmly over the identified spot (just above the umbilicus, slightly left of the midline).
        - Compress the aorta against the spine by applying steady, sustained downward pressure. The force should be strong enough to significantly reduce blood flow to the pelvis.
      - Check for Effectiveness:
        - Observe for a decrease or cessation of vaginal bleeding within seconds, indicating successful compression.
        - Check the femoral pulse (in the groin) to confirm reduced pelvic blood flow—it should weaken or disappear.
        - Monitor the patient's blood pressure, which may improve as blood is redirected to the upper body.
      - Maintain and Monitor:

- Hold the compression continuously until bleeding is controlled, additional interventions are implemented, or the patient is stabilized for transfer.
- If compression is needed for more than 10 minutes, briefly release pressure (for about 30 seconds) every 10 minutes to assess bleeding and prevent lower limb ischemia, then reapply if necessary.

11. Patient Reassessment

- Frequently Reassess the patient as per Universal Care CPG.

Clinical Insights & Additional Information

- Transport all blood-soaked items to the hospital in a biohazard bag.
- Patients must not be left at scene; all postpartum hemorrhage patients require hospital evaluation.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Symptoms such as vaginal bleeding, clots, altered mental status, hypotension, or signs of shock post-delivery.
- Injuries / Information: Obstetric history including gravidity, parity, duration of labor, placenta delivery status, and risk factors (e.g., multiple gestation, previous PPH, or coagulopathies).
- Signs:
  - Vital signs (BP, HR, RR, SpO_2), temperature, mental status (AVPU/GCS).
  - Uterine assessment (fundal height, tone, atony).
  - Perineal/vaginal examination findings (lacerations, hematomas, ongoing bleeding).
  - Estimated blood loss (e.g., number of soaked pads, clot size, or volume >500 mL for vaginal delivery).
  - Signs of hypovolemia or shock (e.g., pallor, clammy skin, tachycardia >100 bpm, hypotension <90/60 mmHg, postural BP drop >20 mmHg).
- Treatment:
  - IV/IO access placement and fluid resuscitation details
  - Oxygen therapy details
  - Tranexamic acid (TXA) administration
  - Fundal massage performed and response (e.g., uterus firming, clot expulsion).
  - External aortic compression, if applied, and duration.
  - Placenta delivery status and management (e.g., awaiting delivery, manual removal not performed pre-hospital).
  - Perineal trauma management (e.g., direct pressure applied).
- Allergies: All known allergies, especially to medications (e.g., TXA, oxytocin).
- Medications: Current or recent use of oxytocin, anticoagulants, or other relevant drugs.
- Background: Chronic illnesses (e.g., hypertension, diabetes), recent hospitalization, or obstetric complications.
- Other:
  - Cause of hemorrhage identified (e.g., atony, trauma, retained tissue, coagulopathy).
  - Transport details, including blood-soaked items in biohazard bag.
  - Dispatch and hospital coordination notes (e.g., notification of PPH and shock status).
  - Document any deterioration or red flag conditions (e.g., uncontrolled bleeding, severe hypotension).

Key Performance Indicators

1. Perform fundal massage for all patients with an atonic uterus.

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